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Paleomagnetism and $^{40}Ar/^{39}Ar$ geochronology of gabbro sills at Mariscal Mountain anticline, southern Big Bend National Park, Texas: Implications for the timing of Laramide tectonism and vertical axis rotations in the southern Cordilleran orogenic belt

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Abstract. Mariscal Mountain anticline is a major Laramide asymmetric fold located along the frontal margin of the Cordilleran orogenic belt in the Trans-Pecos region of southwest Texas. Exposed within the limbs and nose of the anticline are concordant mafic and felsic intrusions whose emplacement, based largely on their arcuate outcrop geometry, has been interpreted to predate folding. Paleomagnetic fold tests on samples from large (≥ 30 m thick) and thin (≤ 1 m thick) gabbro sills are negative at the 99% confidence level. Baked sandstones adjacent to the thick sill give sample directions identical to those of the gabbro, whereas sample directions from a site well removed from thermal effects of the gabbro are scattered. Results of the contact tests are consistent with a thermoremanent origin for the gabbro sill magnetization. The negative fold test indicates that sill intrusion at Mariscal Mountain anticline postdated Laramide folding. Two $^{40}Ar/^{39}Ar$ analyses of plagioclase from the gabbro give discordant age spectra but yield an inverse correlation age of 37.0 ± 1.3 Ma (2σ). This date is similar to those from rocks of similar composition elsewhere in Trans-Pecos Texas, but significantly younger than whole rock and pyroxene K-Ar dates of Late Cretaceous and early Eocene age commonly cited for the Mariscal Mountain gabbro. The paleomagnetic and geochronologic data indicate that Laramide deformation had clearly ceased by late Eocene/early Oligocene time, consistent with absence of contractional deformation in middle Eocene to lower Oligocene volcanic and sedimentary rocks elsewhere in the region. Although paleomagnetic data from the gabbro sills may not adequately average secular variation, the in situ group mean direction (D = 348.6°, I = 50.9°, k = 65.1, α_{95} = 4.3°, n = 18 sites) is indistinguishable from expected mid-Tertiary reference directions. This indicates no evidence of significant vertical axis rotation in the Mariscal Mountain area. Analysis of paleomagnetic data from studies elsewhere in the Big Bend region indicates that significant problems exist with most data sets used to argue for the existence of local vertical axis rotations. Thus we conclude that existing paleomagnetic evidence regarding the timing and magnitude of rotations associated with Basin and Range extension is equivocal.

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1. Introduction

Mariscal Mountain anticline is a major Laramide fold located along the frontal margin of the Cordilleran orogenic belt, in southern Big Bend National Park, Texas (Figure 1). Exposed within the limbs and nose of the anticline are mafic and felsic igneous intrusions thought to be of Late Cretaceous to late Eocene or early Oligocene age (Figure 2). Although largely concordant with the overall geometry of the anticline, the relationship between sill intrusion and folding, as well as the exact age of the mafic sills, has long been a matter of debate [*Stevens*, 1988; *Henry and McDowell*, 1986; *Wilson*, 1972]. Early geologic mapping combined with field observations and isotopic (K-Ar) dating led *Maxwell et al.* [1967] and *Maxwell and Dietrich* [1965] to conclude that sill intrusion predated Laramide folding at Mariscal Mountain anticline; they assigned the sills a latest Cretaceous to early Tertiary age. Subsequent detailed geologic mapping by *Bumgardner* [1976] confirmed this interpretation, although he preferred a middle Eocene age for intrusion. An early or middle Eocene age for these sills, assuming they predated Laramide folding, would indicate that Laramide tectonism in this region extended into at least middle to late Eocene time.

In contrast, *Wilson* [1972] suggested that arguments presented by *Maxwell et al.* [1967] in favor of a prefolding origin were inconclusive, and he raised the possibility that the sills may postdate folding. *Wilson* [1972] noted that concordance of the gabbro sills with Upper Cretaceous strata does not necessarily indicate prefolding intrusion and that postfolding intrusion along bedding planes was a viable alternative hypothesis for sill emplacement. Similarly, *Henry and McDowell* [1986] argued that the field relationships cited as indicative of folding, based largely on the observation that the sill outcrop pattern grossly mimics the anticline geometry, are equivocal and do not require intrusion to predate folding. They also suggested that the isotopic dates presented by *Maxwell and Dietrich* [1965] are unreliable and should be disregarded. They reported a new K-Ar date which indicates a middle to late Eocene age for the sill. This new date, together with the possibility the sill postdates folding, is more in accord with geologic data from igneous intrusions of middle Eocene to Oligocene age elsewhere in Trans-Pecos Texas, all of which appear to postdate Laramide deformation [*Stevens*, 1988].

The purpose of this paper is to use paleomagnetic methods to

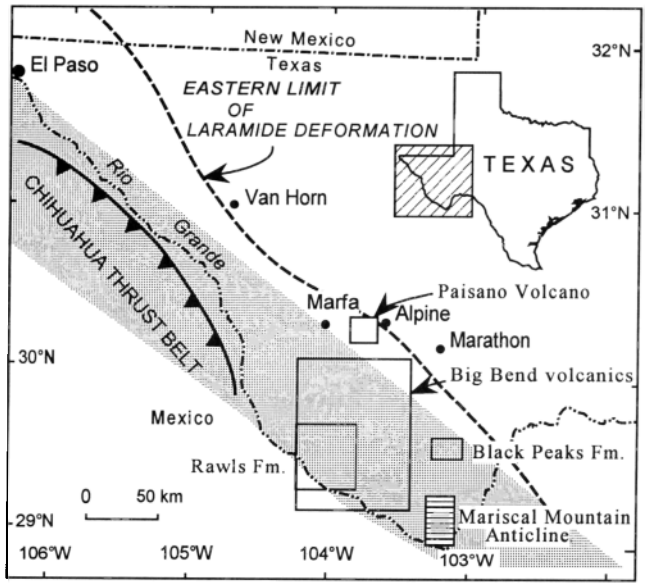


Figure 1. Sketch map of southwestern Texas showing the location of the southern Cordillera orogenic belt and location of Mariscal Mountain anticline (horizontally ruled lines). Also shown are the locations of other paleomagnetic studies discussed in the text and the approximate limit of the Texas Lineament (shaded area) of *Muehlberger and Dickerson* [1989].

test whether or not mafic sill intrusion at Mariscal Mountain anticline predated folding and to report new isotopic age data obtained by the $^{40}Ar/^{39}Ar$ method in order to more precisely determine the age of sill emplacement. Resolution of this relationship is important because of the opportunity to resolve a long-standing geologic controversy and to assess the age of Laramide deformation in this part of the Cordilleran orogenic belt. Finally, paleomagnetic data from this and other published studies are used to assess the extent and magnitude of possible vertical axis rotations associated with Basin and Range extension in west Texas.

2. Geologic Setting

Mariscal Mountain anticline is a narrow north trending, asymmetric, west-vergent fold which can be traced for greater than 15 km (Figure 2). Near its northern termination, the asymmetric geometry dies out into a broad, open, gently north plunging fold (Figure 2). Analysis of structural attitudes from *Maxwell et al.*'s [1967] geologic map of the area, unpublished mapping by J. Oldow and H. Ave Lallemant, and observations made during this study yield a fold axis that plunges 11° in a direction of 342° for the northern part of the fold. Lower Cretaceous strata of the Santa Elena Limestone and Upper Cretaceous (Gulfian) sandstones, shales, and limestones of the Aguja Formation, Pen Formation, Boquillas Formation, Buda Limestone, and Del Rio Clay are deformed within the anticline. The anticline is cut by northwest trending normal faults (Figure 2) that are related to Basin and Range extension but that largely parallel the Laramide structural trends.

Exposed along the limbs and nose of the fold are mafic and felsic, largely concordant intrusions (Figure 2). The largest sill has compositions ranging from alkaline gabbro to syenite and averages 30 m in thickness [*Bumgardner*, 1976]. This sill is present along the western flanks of the fold and within the fold nose and extends several kilometers to the northeast (Figure 2). A sill of similar rock type is present at Cow Heaven anticline, 6 to 10 km northwest of the study area [*Maxwell et al.*, 1967; *Bumgardner*, 1976]. Although the overall geometry of the Mariscal Mountain gabbro is sill-like, contacts are locally discordant and the "sill" cuts up-section approximately 500 m from the Pen Formation in the south to the Aguja Formation in the north [*Bumgardner*, 1976]. This ramp and tread geometry appears to be controlled by the presence of thick competent sandstone or limestone beds and less competent shale layers [*Bumgardner*, 1976]. A similar relationship between mafic intrusions and host sedimentary strata has been reported for Proterozoic diabase sills in the Salt River Canyon area of central Arizona [*Shride*, 1967]. *Bumgardner* [1976] suggested that this thick gabbro sill is actually a series of interconnected, subhorizontal intrusions that grew independently and coalesced into one composite body prior to folding. Also present at Mariscal Mountain, at stratigraphically lower levels, are thin, less than 1 m thick, mafic sills of composition similar to the gabbro described previously.

Felsic sills described as "microgranite" [*Maxwell et al.*, 1967] intrude the Boquillas Formation along the northeastern flank of the anticline (Figure 2). These sills are typically 3 to 10 m thick and parallel sedimentary bedding. They are composed of sparse potassium feldspar, quartz, and plagioclase phenocrysts in a devitrified matrix. Mafic minerals are rare. The absolute age of these sills is not known, but they are thought to be younger than the gabbro sills [*Maxwell et al.*, 1967].

As briefly discussed above, geologic evidence favoring a prefolding origin for the thick gabbro sill is based largely on the concordant nature of the sills and their arcuate geometry that largely mimics that of the anticline (Figure 2). For example, *Bumgardner* [1976, p. 74] states "On a large scale, the entire outcrop pattern of the sill bending around the plunging axis of the anticline supports the hypothesis that the sill was emplaced before the Mariscal Mountain folding." Other arguments involve the presence of late-stage syenitic cylinders and pipe vesicles within gabbro sills. Many of the syenitic cylinders and the vesicles are oriented perpendicular to sill/bedding contacts, although at several localities, syenitic cylinders are randomly oriented. The syenitic cylinders and the pipe vesicles perpendicular to bedding contacts are thought to have risen buoyantly through magma due to a gravitational gradient [*Best*, 1982; *Walker*, 1987a] and are thought to represent the paleohorizontal at the time of sill emplacement. Subsequent deflections of these linear elements from vertical are taken to indicate that the sills have been folded (W.R. Muehlberger, written communication to C.D. Henry and J.G. Price, 1986; D. Potter, personal communication, 1987).

Paleomagnetic data from sills with similar geometry elsewhere in the Cordilleran fold and thrust belt [*Harlan et al.*, 1988] indicate that they clearly postdate the folds in which they have been emplaced. Thus a close correspondence between sill and fold geometry does not provide sufficient evidence to infer the age of folding. Furthermore, the plunge of syenitic cylinders in the flat-lying 28 Ma Rattlesnake Mountain sill [*Carman et al.*, 1975; *Henry and McDowell*, 1986] in northwestern Big Bend National Park varies from vertical to horizontal. Also, recent work on the mechanics of vesicle formation in pillow basalts indicates that

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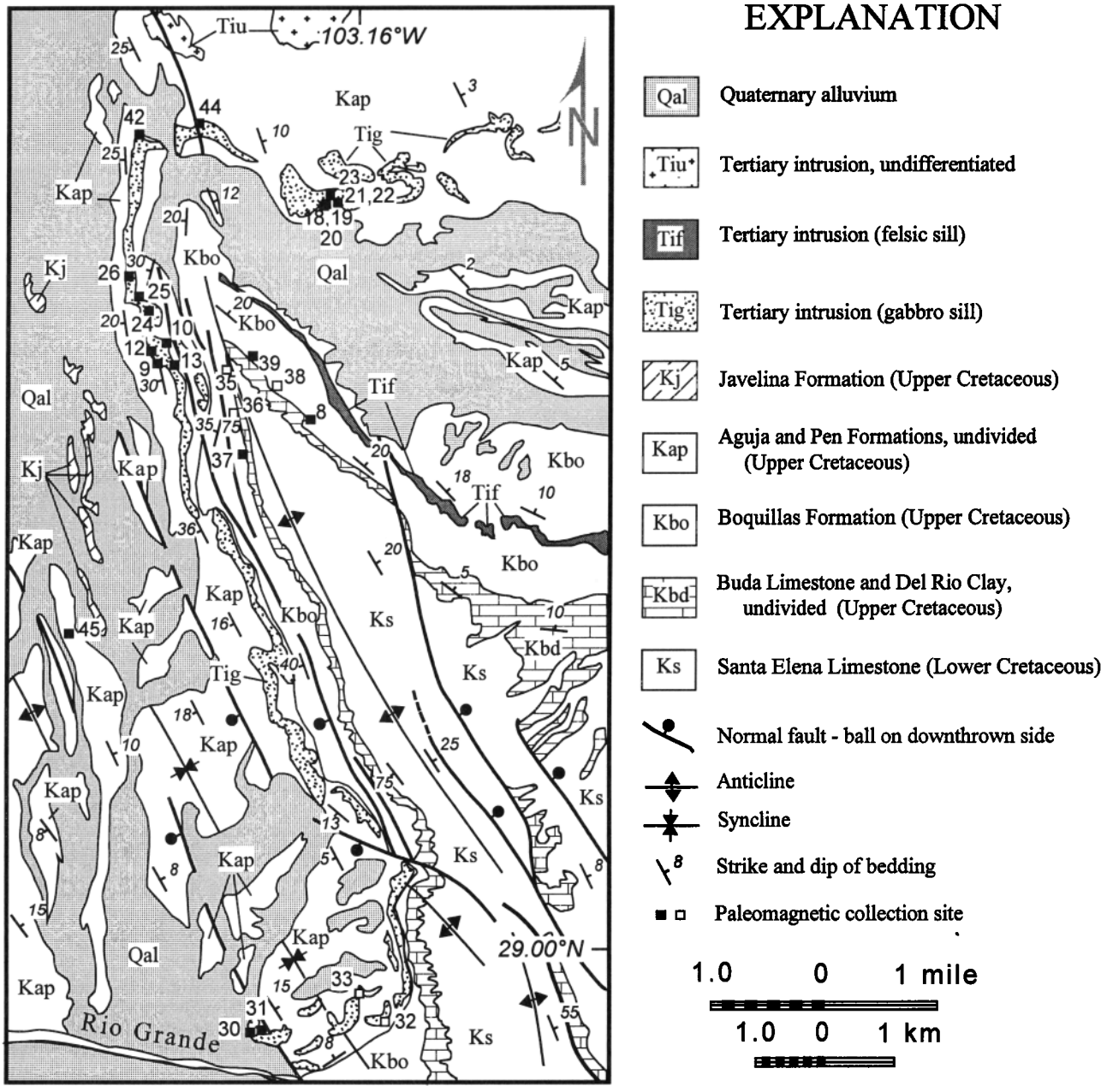


Figure 2. Geologic map of Mariscal Mountain anticline and the location of paleomagnetic sites in gabbro sills and sedimentary rocks sampled in this study. The thin gabbro sills are too small to be shown in this diagram. Open squares are sites that did not yield usable paleomagnetic data. Map is modified from *Maxwell et al.* [1967].

vesicles form as a result of gas exsolution perpendicular to an advancing solidification front in crystallizing magma and may move toward the interior of the body, not toward the margin [*Philpotts and Lewis*, 1987; *Walker*, 1987b]. It appears that pipe vesicles can have any orientation, and neither cylinders nor pipe vesicles may be reliable recorders of paleohorizontal [*Philpotts and Lewis*, 1987]. The field evidence bearing on the age of folding at Mariscal Mountain is clearly equivocal.

Isotopic dating of samples from the gabbro sill at Mariscal Mountain anticline has yielded apparent ages that range from Late Cretaceous to early Oligocene. *Maxwell and Dietrich* [1965], without documentation of analytical results, reported K-Ar dates that ranged from 70 to 50 Ma for a whole rock sample and for an impure pyroxene mineral separate (ages recalculated using the decay constants of *Steiger and Jäger* [1977]). *Henry and McDowell* [1986] in assessing *Maxwell and Dietrich*'s work note that only two analyses were performed and that both the whole rock and pyroxene separate were probably from the same sample. They suggest these data should be disregarded because of 1) the great discordance between the two dates, 2) the altered nature of the gabbro which renders a whole rock age highly questionable, and 3) the impure nature of the pyroxene separate and possibility

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of excess ^{40}Ar in the feldspar or rock fragments that contaminate the concentrate. *Henry and McDowell* [1986] reported a "reliable" K-Ar date of 34.4 ± 3.0 Ma on a pyroxene separate and noted that this date was consistent with numerous 34 Ma dates on mafic intrusive rocks elsewhere in the Big Bend area.

Unfortunately, the gabbro that comprises the thick mafic sill at Mariscal Mountain anticline is a poor material for K-Ar dating. The gabbro is composed primarily of pyroxene and plagioclase, both phases low in potassium. Biotite and hornblende are not present. Both pyroxene and plagioclase are highly altered, and *Bumgardner* [1976] reported finding no samples adequate for whole rock geochemical analysis. Both pyroxene and plagioclase are dubious candidates for K-Ar dating as both commonly contain excess ^{40}Ar. Thus, it is questionable if any of the K-Ar dates from the sill can be considered reliable in the absence of confirmation by other methods.

3. Field and Laboratory Procedures

Five to ten independently oriented samples were collected from 23 sites located in gabbro sills that intrude sedimentary strata at Mariscal Mountain anticline (Figure 2). Six sites were collected from thin gabbro sills that intrude the Boquillas Formation and Buda Limestone; the rest were from the thick gabbro sill which cuts the Pen and Aguja Formations. Samples from four sites were collected from Upper Cretaceous sedimentary strata adjacent to mafic sills in order to conduct baked contact tests [*Everitt and Clegg*, 1962]. A site located in the Cretaceous Aguja Formation well removed from the sills (Figure 2) was sampled in order to test for the presence of possible widespread tectonothermal remagnetization of rocks in the Mariscal Mountain area which could reflect widespread igneous activity subsequent to sill emplacement. Finally, nine sites were sampled in the felsic sill and adjacent strata; results obtained during laboratory analysis were dominantly of reverse polarity, but demagnetization results were commonly erratic. Data from these sites will not be considered further.

Samples were collected using a portable drill and were oriented using magnetic and sun compasses. Collection sites were chosen such that the attitude of adjacent sedimentary strata could be measured for the application of tilt corrections. In a few cases reliable attitudes could not be obtained because adjacent sedimentary strata were poorly exposed. For these sites, structural attitudes were extrapolated from *Maxwell et al.*'s [1967] geologic map. Sites were also located to provide as wide a range of structural attitudes as possible for the fold test.

Measurements of the natural remanent magnetization (NRM) of individual specimens were made using either a cryogenic or spinner magnetometer. Progressive stepwise demagnetization was conducted on all samples by either alternating field (AF) or thermal methods. Magnetization directions were determined using principal component analysis [*Kirschvink*, 1980]. Only magnetizations defined by four or more colinear points that decayed toward the origin of vector diagrams and that represented a significant demagnetization interval (i.e., ≥ 15 to 20 mT) were used in the analysis. All sample directions used in the calculation of site mean directions had maximum angular deviation (MAD) values less than 8°. Identification of magnetic carriers was made by using saturation magnetization versus temperature (Curie temperature) curves, by acquisition of isothermal remanent magnetization (IRM), and by AF and direct current (d.c.) demagnetization of IRM techniques.

For isotopic dating, a plagioclase separate was extracted from a sample of the porphyritic border phase of the thick gabbro sill by crushing and magnetic separation. Because the plagioclase contained some zeolite (natrolite) alteration along fractures, the separate was divided, and one split was leached with hot HNO_3 (8436TT) for 30 min in order to remove the zeolite. The other (8436T) was leached with hot HNO_3 for 30 min followed by dilute (5%) HF treatment for 30 min. Neither treatment was fully successful in removing the contaminating phase. The plagioclase separates were irradiated for 20 hours in the reactor facility of McMaster University, Hamilton, Ontario. Neutron fluence was monitored using hornblende standard MMhb-1, which has a K-Ar age of 520.4 ± 1.7 Ma [*Samson and Alexander*, 1987]. After irradiation, the samples and standards were degassed in a silica glass tube using a Lindberg resistance furnace following the procedure of *Onstott and Peacock* [1987]. The errors in isotopic ratios and the apparent age for each heating step were determined by the procedure described by *Onstott and Peacock* [1987]. The data were also corrected for radioactive decay of ^{36}Cl, ^{37}Ar, and ^{39}Ar and isotopic interferences from K, Ca, and Cl using correction factors and errors given by *Onstott and Peacock* [1987].

The $^{40}Ar/^{39}Ar$ data are plotted as apparent age spectra, and the combined data from both samples are represented on an inverse correlation diagram. For the correlation diagram, the apparent age and its error were calculated from the intercept of the best fit line [*York*, 1969] with the $^{39}Ar/^{40}Ar$ axis. The error estimates for the intercept date have been multiplied by the square root of the mean square of weighted deviates (MSWD) in order to account for scatter of the data points about the best fit line. The errors quoted for the integrated and $^{39}Ar/^{40}Ar$ intercept dates incorporate errors in the flux measurement, interference corrections, and age of the standard.

4. Results

4.1. Paleomagnetism of the Gabbro Sills

Analysis of paleomagnetic data from most gabbro sills was relatively straightforward. NRM intensities ranged from 0.0033 A/m to 110 A/m, with a geometric mean of 1.75 A/m. For most samples, a single remanent magnetization component, characterized by north declinations (D) and moderate positive inclinations (I), was identified by the univectorial decay of the magnetization to the origin of a demagnetization diagram during either AF or thermal demagnetization (Figures 3a and 3b). These samples were characterized by moderate to high median destructive fields in AF demagnetization (15-30 mT) and by thermally discrete or distributed laboratory unblocking temperatures below 580°C during thermal demagnetization (Figures 3a and 3b). We believe that the remanence isolated in these samples is probably a thermoremanent magnetization (TRM) acquired during cooling following emplacement of the gabbro sills.

Some samples displayed slightly more complicated behavior during demagnetization experiments. These samples are characterized by the presence of a minor low coercivity component with due north or northeast declination. This magnetization was usually removed by AF demagnetization to 10 or 20 mT or by temperatures of less than 250° to 300°C, after which the characteristic remanence was isolated (Figures 3c and 3d). This secon-

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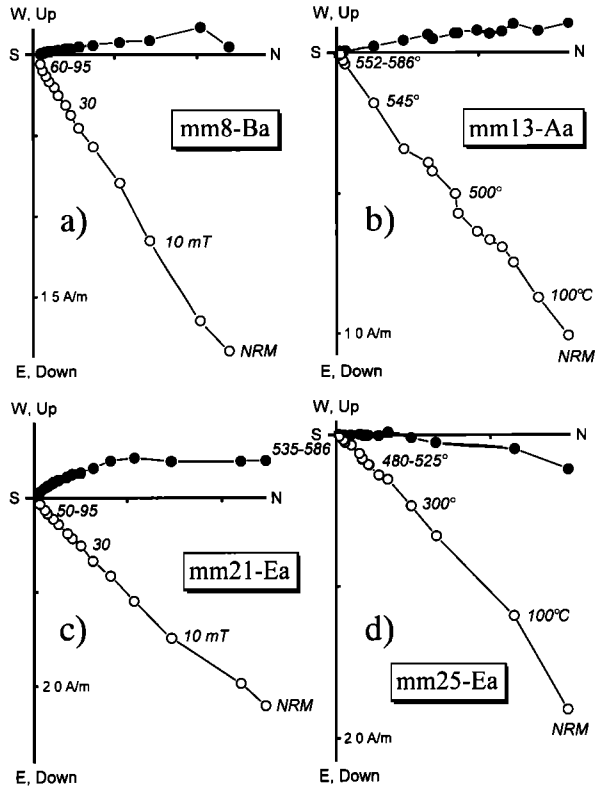


Figure 3. Orthogonal vector diagrams showing representative alternating field and thermal demagnetization behavior of gabbro sills from Mariscal Mountain anticline. Solid and open symbols represent the in situ magnetization components projected onto the horizontal and vertical planes, respectively. Demagnetization is in milliteslas (mT) and degrees Celsius.

dary component is probably a viscous remanent magnetization acquired in the present-day field that is superimposed upon the primary sill remanence.

Magnetization directions isolated by principal component analysis generally had MAD values less than 5°. Directional data from individual sites are, for the most part, well defined as indicated by α_{95} values that ranged from 2.1° to 13.8°, with 77% having an $\alpha_{95} \leq 5.0°$. Precision parameters (k) for most sites [*Fisher*, 1953] exceeded 100 (Table 1).

Anomalous paleomagnetic behavior was observed at several sites. NRM intensities from individual samples typically ranged from 10 to 110 A/m. AF demagnetization of these samples yielded extremely well defined (e.g., MAD values < 2.0°), low-coercivity magnetizations. Within-site scatter, however, was extremely high. As many of these sites were located in topographically high or exposed areas, we suggest that these samples record an isothermal remanent magnetization that was probably caused by lightning strikes. Five sites which displayed such behavior were rejected from statistical analysis, and their site mean directions have been excluded from Table 1.

In IRM experiments, the gabbro samples are characterized by rapid acquisition of IRM in applied D.C. fields, and all samples achieved complete saturation in fields less than 200 mT. This behavior is consistent with a remanence dominated by a low coercivity magnetic phase. Unblocking of magnetization in laboratory temperatures less than 580°C and reversible thermomagnetic curves with Curie temperatures of 571° ± 5°C (n = 4 samples) indicate the presence of low-Ti titanomagnetite. Rock magnetic behavior demonstrates that magnetite is the primary carrier of magnetization in gabbro samples from Mariscal Mountain anticline.

4.2. Baked Contact Test

Samples from contact sandstones and mudstones in the thermally metamorphosed zone adjacent to the main gabbro sill (sites mm19, mm42b, and mm22; Figures 4a-4c) have NRM intensities that range from 0.00089 A/m to 5.52 A/m, with a geometric mean of 0.035 A/m. These samples typically yield well-defined (≤ 5° MAD values), single-component magnetizations of moderate to high median destructive fields (20 to 40 mT) and a relatively narrow range of laboratory unblocking temperatures (> 50 to 80% of unblocking occurring between 500° to 570°C) (Figures 4a-4c). Site mean directions obtained are statistically indistinguishable from those of adjacent gabbro samples (Figures 5a-5c). However, samples from the contact zone marginal to the steep, west dipping, thin (less than 1 m thick) sill at site mm39 do not yield a remanence similar to that obtained for the intrusion (Figure 5d). Samples from site mm45, located in the Aguja Formation approximately 2.5 km from the nearest exposure of Mariscal Mountain sills, are characterized by low NRM intensities (geometric mean = 0.00056 A/m), low median destructive fields (5 to 10 mT), and low (250° to 550°C) laboratory unblocking temperatures (Figure 4d). The magnetizations are generally poorly defined (MAD values 10° to 20°) relative to samples from the baked contacts of the large sills. Furthermore, directions obtained from these samples are highly scattered (Figure 5e), and within-sample consistency between results obtained by thermal or AF methods is poor. Paleomagnetic and rock magnetic data from baked and unbaked sandstones indicate that the remanence is carried by magnetite.

4.3. Paleomagnetic Fold Test

Three data sets were subjected to paleomagnetic fold tests [*Graham*, 1949] to determine whether sill intrusion predated or postdated folding at Mariscal Mountain anticline. The first data set involves comparison of in situ and structurally corrected site mean directions from sites in both thick and thin gabbro sills. The second set includes only sites from the thick sill, and the third involves only results from sites in the thin sills. Structural corrections for the site mean directions were applied by rotating adjacent host strata to horizontal about the line of bedding strike. Although the fold plunges shallowly northwest, corrections for plunge were not applied as the error introduced by ignoring plunge is probably negligible for folds that plunge 10° or less [*MacDonald*, 1980]. Because several sites are closely spaced and have identical structural corrections (Table 1), their site mean directions have been averaged; these are treated as single observations in the fold tests described below. These include sites mm18 and 20 with $D = 327.9°$, $I = 45.9°$, sites mm21 and 23 with $D = 337.8°$, $I = 44.8°$, and sites mm24, mm25, and mm26 with $D = 0.5°$, $I = 47.5°$.

The statistical level of significance of the fold tests was determined using methods described by *McElhinny* [1964] and

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Table 1. Summary of Site Mean Paleomagnetic Data From Mariscal Mountain Anticline, Southern Big Bend National Park, Texas

Site	*S*lat	*S*long	n/n_o	D_{is}	I_{is}	D_{sc}	I_{sc}	Attitude	k	α_{95}	*P*lat	*P*long	Rock Type
mm8	29.085°N	103.177°W	10/10	355.6°	53.3°	15.1°	43.0°	335°/20°	201.9	3.4°	83.9°N	219.7°E	thin gabbro sill
mm9	29.087°N	103.196°W	5/7	345.7°	45.1°	313.6°	38.3°	160°/35°	31.7	13.8°	77.1°N	159.3°E	thick gabbro sill
mm10	29.087°N	103.195°W	7/7	358.7°	50.4°	305.3°	59.1°	140°/36°	316.1	3.4°	87.7°N	228.5°E	thick gabbro sill
mm12	29.085°N	103.193°W	9/10	345.5°	56.0°	295.8°	49.3°	150°/36°	246.1	3.3°	75.7°N	202.1°E	thick gabbro sill
mm13	29.086°N	103.192°W	7/8	351.4°	54.2°	314.0°	47.3°	165°/30°	870.7	2.1°	80.8°N	206.7°E	thick gabbro sill
mm18	29.106°N	103.171°W	5/8	331.1°	40.2°	334.0°	31.0°	266°/10°	335.6	4.2°	63.4°N	160.1°E	thick gabbro sill
mm19*	29.106°N	103.171°W	6/6	323.7°	42.2°	327.6°	33.6°	266°/10°	186.1	4.9°	57.4°N	167.2°E	contact sandstone
mm20	29.106°N	103.171°W	8/8	323.9°	51.4°	329.2°	42.7°	266°/10°	468.6	2.6°	58.9°N	181.5°E	thick gabbro sill
mm21	29.108°N	103.171°W	8/8	331.2°	44.3°	333.7°	34.8°	260°/10°	123.0	5.0°	64.5°N	166.7°E	thick gabbro sill
mm22*	29.108°N	103.171°W	8/8	320.6°	47.2°	324.8°	38.3°	260°/10°	131.2	4.9°	55.6°N	175.4°E	contact sandstone
mm23	29.109°N	103.172°W	9/9	344.4°	45.0°	345.2°	35.0°	260°/10°	167.0	4.0°	76.0°N	160.1°E	thick gabbro sill
mm24	29.092°N	103.196°W	8/9	9.8°	42.1°	351.1°	33.5°	200°/25°	158.6	4.4°	80.0°N	13.2°E	thick gabbro sill
mm25	29.094°N	103.197°W	9/10	358.2°	48.4°	338.8°	35.0°	200°/25°	70.6	6.2°	88.4°N	177.7°E	thick gabbro sill
mm26	29.099°N	103.201°W	10/10	352.1°	51.2°	332.9°	35.6°	200°/25°	449.5	2.3°	82.6°N	191.0°E	thick gabbro sill
mm30	28.987°N	103.178°W	9/9	5.7°	55.8°	19.5°	45.7°	331°/15°	179.3	3.9°	81.2°N	288.4°E	thick gabbro sill
mm31	28.988°N	103.177°W	7/7	347.0°	50.1°	7.1°	46.6°	346°/18°	271.2	3.7°	78.6°N	179.6°E	thick gabbro sill
mm32†	28.988°N	103.166°W	0/8										thick gabbro sill
mm33†	28.987°N	103.167°W	0/9										thick gabbro sill
mm35†	29.088°N	103.181°W	0/9										thin gabbro sill
mm36†	29.085°N	103.187°W	0/8										thin gabbro sill
mm37	29.084°N	103.187°W	9/10	4.1°	55.3°	297.6°	30.9°	168°/63°	122.9	4.6°	82.4°N	282.9°E	thin gabbro sill
mm38†	29.086°N	103.181°W	0/10										thin gabbro sill
mm39	29.089°N	103.184°W	7/8	1.8°	54.2°	30.2°	51.9°	10°/21°	118.1	5.6°	84.2°N	271.5°E	thin gabbro sill
mm42a	29.116°N	103.201°W	9/10	348.3°	53.2°	349.9°	33.3°	264°/20°	268.9	3.2°	79.0°N	194.7°E	thick gabbro sill
mm42b*	29.116°N	103.201°W	4/4	344.5°	54.5°	347.3°	34.7°	264°/20°	95.6	7.2	75.6°N	195.0°E	contact sandstone
mm44	29.118°N	103.192°W	8/9	325.9°	51.4°	334.0°	35.4°	268°/18°	56.5	8.2°	60.6°N	181.2°E	thick gabbro sill
mm45*	29.042°N	103.211°W	10/14	3.7°	52.6°	7.3°	51.5°	345°/3°	18.8	11.5°	83.0°N	318.6°E‡	unbaked sandstone

*S*lat and *S*long are latitude and longitude of the paleomagnetic collection site; n/n_o is ratio of samples used in the calculation of the site mean direction to the total number of samples collected and measured; D_{is} and I_{is} are declination and inclination of the in situ site mean direction; D_{sc} and I_{sc} are declination and inclination of the site mean direction after application of tilt corrections; attitude is strike/dip; k is precision parameter [*Fisher*, 1953]; α_{95} is semi-angle of the cone of 95% confidence about the site mean direction; *P*lat and *P*long are the latitude and longitude of the in situ virtual geomagnetic pole. Structurally corrected site mean directions were restored to horizontal by untilting the direction about the line of strike of bedding. For the attitude, the direction of dip is measured 90° clockwise from strike.

*site was not used in the calculation of the group mean direction.

†Samples from this site were unstable during demagnetization or yielded grossly inconsistent sample directions, and a site mean direction was not calculated.

‡Pole position is based on the structurally corrected mean direction.

McFadden [1990]. McElhinny's test uses the ratio of *Fisher's* [1953] precision parameter k for both the in situ and structurally corrected group mean directions. This ratio is compared against the F distribution for the appropriate sample size (n) and desired level of significance (usually 0.05). If the observed ratio exceeds the test statistic, the test is significant at that level. This test assumes that both the in situ and structurally corrected populations are Fisherian distributions, a condition usually violated by the effects of folding [*McFadden and Jones*, 1981; *McFadden*, 1990]. This test, although not strictly valid, is typically very conservative with the result that if a fold test is significant using the *McElhinny* [1964] test, then it usually satisfies the more stringent criteria of the McFadden test described below [*McFadden and Jones*, 1981; *McFadden*, 1990]. The *McElhinny* [1964] test continues to be used by many paleomagnetists, however, largely because of its ease of use and the intuitive recognition derived from visual comparison of in situ and structurally corrected precision parameters. Despite the limitations described above, we have used *McElhinny's* [1964] test, as well as the more appropriate *McFadden* [1990] test described below.

Several alternatives to the *McElhinny* [1964] statistical method have been proposed to account for complexities in paleomagnetic sampling, fold geometry, and statistical formulations [*McFadden and Jones*, 1981; *McFadden*, 1990; *Watson and Enkin*, 1993; *Tauxe and Watson*, 1994]. The test used here is based on a correlation between site mean directions of in situ and structurally corrected orientations with their structural corrections using the test statistic ξ [*McFadden*, 1990, equation 14]. This test is sensitive to a correlation between the distribution of site mean directions about the overall group mean, and the tectonic correction. If ξ is large, a correlation exists between the distribution of site means and tectonic corrections, and thus it is unlikely that the observed magnetization was acquired with the sites in their present relative position. Like the *McElhinny* [1964] test, the observed test statistic is compared with the critical value for a given n to determine the appropriate level of significance. This test is highly flexible and requires only about 40% of the tectonic movement that the *McElhinny* [1964] test requires to achieve the same level of significance. Although both the *McElhinny* [1964] and the *McFadden* [1990] statistics are reported here, greater weight is given to the results of the correlation test of *McFadden* [1990].

In situ results from the large and small gabbro sills are well

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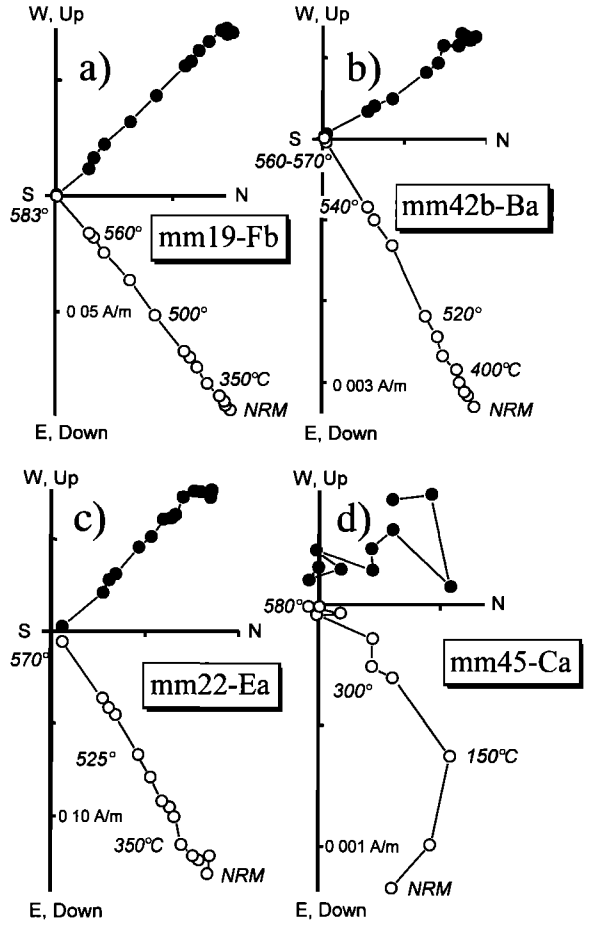


Figure 4. Orthogonal vector diagrams showing the response of (a-c) baked and (d)unbaked sandstones to thermal demagnetization. Conventions are as in Figure 3.

grouped and yield a mean direction of $D = 349.2°$, $I = 51.9°$, with $k_{is} = 83.9$, and $\alpha_{95} = 4.4°$ ($n = 14$ independent observations) (Figure 6a; Table 2). Application of tilt corrections results in a pronounced divergence of site mean directions (Figure 6b) and a decrease in the value of k and increase of α_{95}. The mean direction and statistics after structural correction are $D = 337.7°$, $I = 45.7°$, $k_{sc} = 12.5$, and $\alpha_{95} = 11.7°$ (Table 2). The test statistic $k_{is}/k_{sc} = 6.71$, which exceeds the critical value required at the 99% confidence level (i.e., 2.55; Table 2) for the *McElhinny* [1964] test; this result indicates that the fold test is negative at the 99% confidence level. Using the correlation test of *McFadden* [1990], ξ_{is} for the data set of in situ mean directions is 1.72, which does not exceed the critical value of 6.09 at the 99% confidence level (Table 2). There is no apparent correlation between in situ site mean directions and the structural corrections. For the structurally corrected mean, ξ_{sc} is 11.30, which exceeds the test statistic indicating that a correlation exists between the site mean directions and tectonic corrections. This result indicates that it is unlikely that the sill remanence was acquired prior to folding, and thus the fold test is negative.

Essentially identical results are obtained by using only sites from the thick gabbro sill. For the *McElhinny* [1964] test k_{is}/k_{sc}, is 4.71, whereas the critical value at 99% confidence is 2.94 (Table 2). Again, this indicates a negative fold test. For the *McFadden* [1990] test, ξ_{is} is 2.31 and ξ_{sc} is 8.58. The test statistic ξ_{crit} is 5.38 at the 99% confidence level. This indicates that there is no correspondence between the in situ site means and structural corrections, but that significant correlation exists between the structurally corrected site mean directions and structural corrections. Put simply, this indicates that the fold test is negative and that the remanence was acquired after the fold developed.

Finally, we have applied a fold test to the thin gabbro sills. Figure 6b shows that application of structural corrections result in a dramatic divergence of the three site mean directions, and this is reflected by their respective k values ($k_{is} = 860.9$, $k_{sc} = 4.8$) (Table 2). The k_{is}/k_{sc} value for the *McElhinny* [1964] fold test is 179.4, considerably greater than the 16.0 required for significance at the 99% confidence level. For the *McFadden* [1990] test $\xi_{is} = 0.498$ and $\xi_{sc} = 3.00$, the ξ_{crit} value is 2.662. Again, these fold tests are negative at the 99% level.

Statistical analysis of in situ and structurally corrected results, regardless of the statistical method used or whether sites from the large and small sills are used, indicates that the fold tests are negative at the 99% confidence level. Paleomagnetic data indicate that the characteristic magnetization of the gabbro sills postdates folding at Mariscal Mountain anticline, and we conclude that the gabbro sills were emplaced after Laramide folding.

4.4. Isotopic Dating

The two plagioclase separates from the Mariscal Mountain gabbro yield discordant $^{40}Ar/^{39}Ar$ age spectra (Figure 7) and integrated or "total gas" dates of 43.2 ± 1.1 and 35.5 ± 1.2 Ma (2σ) for 8436TT and 8436T, respectively (Table 3). The $^{40}Ar/^{39}Ar$ step-heating spectrum for 8436TT rises from 35 Ma at low-temperature steps to a maximum value of 58 Ma at intermediate steps, before dropping to 30 Ma at higher temperatures. The age spectrum for 8436T, however, is much flatter, yielding dates which range from 30 to 40 Ma (Figure 7). The $^{37}Ar_{Ca}/^{39}Ar_K$ values for 8436T vary between 5 and 15 (Ca/K = 9 to 27). The $^{37}Ar_{Ca}/^{39}Ar_K$ values for 8436TT, however, fall within this range for only the low- and high-temperature steps (Figure 7), whereas intermediate temperature steps yield extremely low values of 0.3 to 0.7 (Ca/K = 0.5 to 1.3). The oldest dates are derived from these same intermediate temperature steps (Figure 7). Because the $^{39}Ar_K$ content for 8436TT is higher than that of 8436T, yielding estimated percent K of 0.88 and 0.22, respectively, the low $^{37}Ar_{Ca}/^{39}Ar_K$ values for 8436TT (Figure 7) must be due to contamination by a K-rich phase, such as zeolite, which is not as abundant in 8436T because it was greatly reduced by the HF treatment. The presence of excess argon in this K-rich phase would also account for the anomalously high dates in the intermediate part of the 8436TT spectrum.

When data from both samples are combined on a correlation diagram (Figure 8), 21 of the 24 analyses fall about a mixing line between atmospheric contamination (i.e., $^{36}Ar/^{40}Ar = 0.003356$ which yields an initial $^{40}Ar/^{36}Ar$ ratio of 298 ± 3) and $^{39}Ar/^{40}Ar$ intercept of 0.1980 which corresponds to an age of 37.0 ± 1.3 Ma (2σ). The MSWD value of 15 indicates that the data are more scattered than expected from analytical uncertainties, which is consistent with the irregularity in the age spectra. Three points lying to the left of the mixing line correspond to the excessively old dates of the 8436TT spectrum and have been excluded from the regression as they probably contain excess radiogenic ^{40}Ar.

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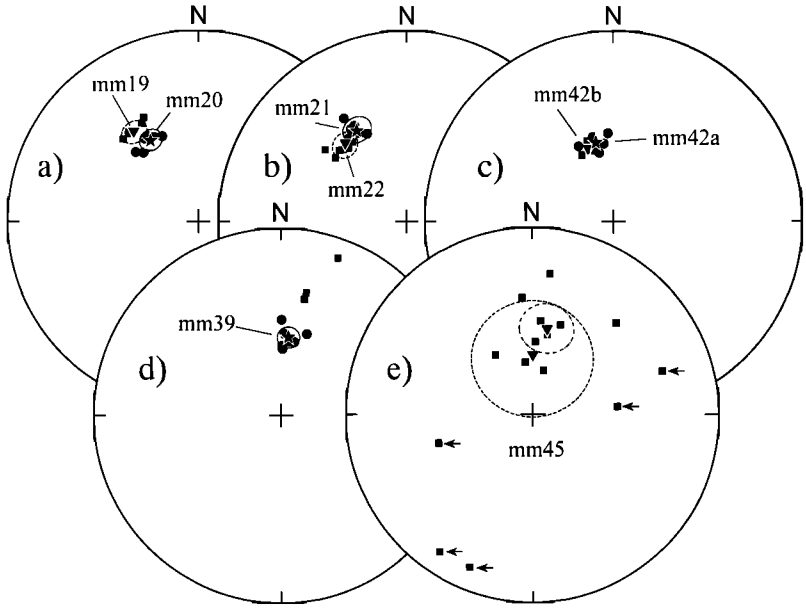


Figure 5. Equal-area projections showing the results of paleomagnetic contact tests from (a-c) thick and (d) thin gabbro sills and (e) paleomagnetic results from sandstone of the Aguja Formation, well removed from thermal effects of the Mariscal Mountain gabbro sills. Squares represent sample directions from sandstones, circles are directions from gabbros, downward pointing triangles are site mean directions calculated for the sandstones in Figures 5a-5c and 5e, and stars are site mean directions from gabbro. Solid lines are α_{95} cones of confidence for gabbro site mean directions; dashed lines are from the sandstones. In Figure 5e the dashed α_{95} line represents the site mean direction calculated using all sample directions; the dot-dashed line indicates the mean after excluding outlying points denoted by the arrows. The α_{95} for the gabbro and sandstone means has been omitted from Figure 5c for clarity. Figures 5a-5d are in situ coordinates; Figure 5e has been corrected for bedding tilt.

Alternatively, these points may reflect recoil distribution of $^{39}Ar_K$ from the K-rich inclusion phase into the plagioclase, but the differences between the two integrated dates are more consistent with excess ^{40}Ar contamination. We consider the inverse correlation age of 37 Ma a reasonable estimate for the age of the Mariscal Mountain gabbro. This date and its analytical uncertainty span the late Eocene/early Oligocene boundary at 36.6 Ma, using the timescale of *Berggren et al.* [1985]. We note that this age is similar to dates from rocks of similar composition elsewhere in Trans-Pecos Texas [*Henry and McDowell*, 1986].

5. Discussion

5.1. Relationship of Sill Emplacement to Folding at Mariscal Mountain Anticline

Paleomagnetic data from the gabbro sills at Mariscal Mountain anticline yield negative fold tests that are significant at the 99% confidence level; remanence acquisition clearly postdated folding. Of critical concern to our study is whether the observed postfolding remanence is due to cooling of the sills following intrusion or reflects an unrecognized pervasive thermal overprint due to widespread middle Tertiary igneous activity [*Henry and McDowell*, 1986]. The abundance of nearby Tertiary intrusive and volcanic centers prompts suspicion as to whether the entire region may be floored at shallow depths by an intrusion of batholithic dimensions. To test this possibility, paleomagnetic data from samples from the contact zone in Upper Cretaceous strata at several sites were examined. Samples located within 5 m of the thick sill yielded well-defined magnetizations statistically indistinguishable at 95% confidence from magnetizations from the adjacent gabbro suggesting a positive contact test (Figures 5a-5c). Unfortunately, where sampled, these strata are only slightly tilted, and the observed remanence is also similar to that expected for Upper Cretaceous strata in the Big Bend area. If pervasive remagnetization had occurred, a close correspondence between magnetizations in the gabbro and sedimentary strata would not be unreasonable. Results from a contact test adjacent to a thin sill, however, do not yield a magnetization direction similar to that of the gabbros or baked sandstones next to the thick gabbro sill (Figure 5d). Apparently this sill, less than a meter thick, did not thermally remagnetize the adjacent strata. Overall, the contact tests from the thick sills are inconclusive because we cannot demonstrate that the gabbro remanence has been superimposed upon a preexisting magnetization in the sedimentary rocks, but the absence of an identical magnetization in the thin sill indicates that widespread remagnetization did not occur. Furthermore, sedimentary strata well removed from thermal effects associated with gabbro sills at Mariscal Mountain (mm45; Table 1, Figure 5e) yield magnetizations that are widely scattered in direction and display rock magnetic properties distinctly different (i.e., low coercivities in AF demagnetization and low unblocking temperatures for unbaked rocks versus medium to high coercivities and moderate to high unblocking temperatures for baked samples) than those typical of

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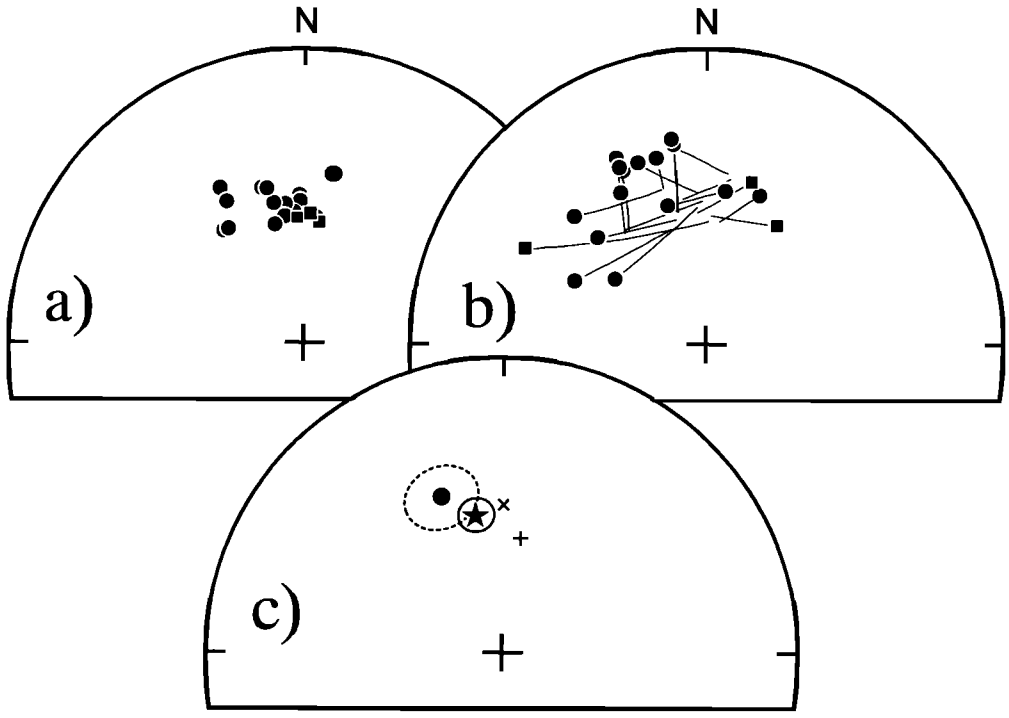


Figure 6. Equal-area projections showing paleomagnetic results from gabbro sills at Mariscal Mountain anticline and the effects of the paleomagnetic fold test: (a) in situ site mean directions (n = 18 sites), (b) site mean directions after untilting based on the attitude of adjacent sedimentary strata (n = 18 sites), and (c) in situ (star) and structurally corrected (solid circle) site mean directions based on the 14 independent observations used in the fold test (see text) and their α_{95} cones of confidence. The cross in Figure 6c is the present axial geocentric dipole direction for the study area, and the plus is the direction of the present-day field. Circles in Figures 6a and 6b are site mean directions from thick gabbro sills; squares are from thin gabbro sills. The lines in Figure 6b connect the location of the in situ direction and structurally corrected directions, and show the path of paleomagnetic direction as the host strata are restored to horizontal.

samples from sites adjacent to the large gabbro sill (Figures 5e and 4d). These observations suggest that the contact tests for the large sills are positive and that the observed postfolding remanence characteristic of the sills is probably a TRM acquired during cooling subsequent to emplacement. We conclude that sill intrusion clearly postdated Laramide folding at Mariscal Mountain anticline.

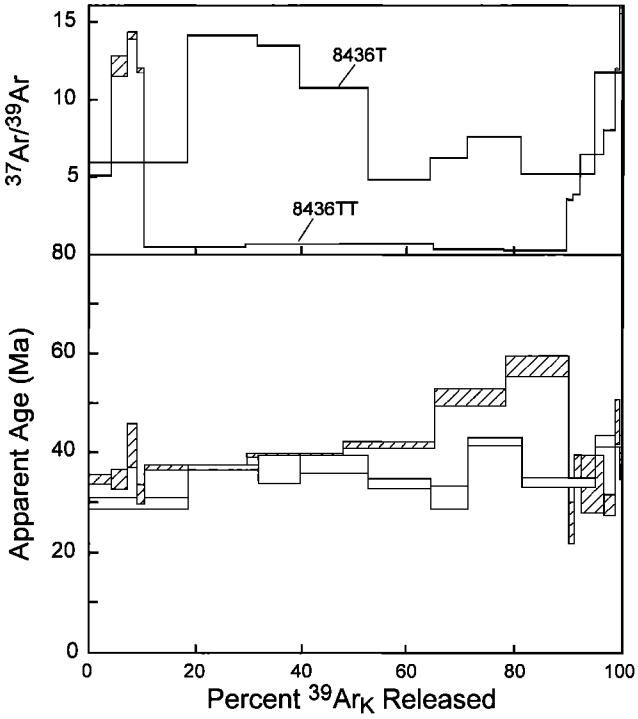
The observed somewhat elongate distribution of in situ site mean directions (Figure 6a) may be due to inadequate averaging of secular variation. This is a distinct possibility given the limited thickness of the sills and relatively short time required to cool a 30-m-thick sill at shallow crustal levels. Because both thin and thick sills were sampled, we suggest that at least two independent recordings of the geomagnetic field are recorded by the gabbros at

Table 2. In Situ and Structurally Corrected Paleomagnetic Group Mean Directions From the Gabbro Sills at Mariscal Mountain Anticline, Southern Big Bend National Park, Texas

In Situ					Structurally Corrected					Test Statistics		
D	I	k_{is}	ξ_{is}	α_{95}	D	I	k_{sc}	ξ_{sc}	α_{95}	k_{is}/k_{sc}	k_{crit}	ξ_{crit}
Fold Test 1—All Sites in the Gabbro Sills (n = 14 Observations)												
349.2°	51.9°	83.9	1.72	4.4°	337.7°	45.7°	12.5	11.30	11.7°	6.71	2.55	6.09
Fold Test 2—Sites in Thick Sill Only (n = 11 Observations)												
346.4°	51.0°	81.4	2.31	5.1°	334.7°	44.4°	17.3	8.58	11.3°	4.71	2.94	5.38
Fold Test 3—Sites in Thin Sills (n = 3 Observations)												
0.4°	54.3°	860.9	0.498	4.2°	351.0°	50.1°	4.8	3.00	63.8°	179.4	16.0	2.66

D and I are declination and inclination of the group mean direction; k_{is} and k_{sc} are precision parameter [*Fisher*, 1953] of the in situ and structurally corrected mean directions; ξ_{is} and ξ_{sc} are observed value of the test statistic for directional correspondence [*McFadden*, 1990] for the in situ and structurally corrected mean directions; α_{95} is cone of 95% confidence about the group mean direction; k_{is}/k_{sc} is ratio of the in situ k value to the structurally corrected k value; k_{crit} is value of the test statistic that must be exceeded to satisfy the null hypothesis at the 99% confidence level for the *McElhinny* [1964] fold test; ξ_{crit} is the value of the test statistic that must be exceeded to satisfy the null hypothesis at the 99% confidence level for the *McFadden* [1990] fold test.

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Mariscal Mountain. If the main gabbro sill was formed by independent injection of multiple pulses of magma over a limited period of time as suggested by *Bumgardner* [1976], then at least some averaging of secular variation may be expected. Some dispersion in site mean directions may also be due to deformation caused by injection of adjacent sill segments or to subsequent small-magnitude normal faulting. We note that the observed in situ grand mean direction for the 18 sites ($D = 348.6°$, $I = 50.9°$, $k = 65.1$, $\alpha_{95} = 4.3°$; Table 4) is, however, statistically indistinguishable at the 95% confidence level from expected reference directions calculated from either *Harrison and Lindh's* [1982] 40 Ma pole or *Diehl et al.'s* [1988] mid-Tertiary (20 to 40 Ma) pole (Figure 9a).

Paleomagnetic and isotopic data from the Mariscal Mountain gabbro sills indicate that Laramide folding in the Big Bend area had ceased prior to late Eocene time. This conclusion is consistent with an absence of contractional deformation in the middle Eocene Canoe Formation and younger sedimentary strata in Big

Figure 7. $^{40}Ar/^{39}Ar$ apparent age and $^{37}Ar/^{39}Ar$ spectra for plagioclase samples from the Mariscal Mountain gabbro. Width of the horizontal bar in each diagram represents the analytical error at 1σ. Open error bars are from sample 8436T; diagonally ruled errors are from 8436TT.

Table 3. $^{40}Ar/^{39}Ar$ Data From Plagioclase Separates From the Mariscal Mountain Gabbro, Southern Big Bend National Park, Texas

Temperature °C	$f1$	$f2$	$\%^{40}Ar_R$	^{40}Ar $x10^{-6}cm^3STP$	$\%^{39}Ar$	$^{37}Ar/^{39}Ar$	$^{38}Ar/^{39}Ar$	Apparent Age Ma ± 1σ
Mariscal Mountain plagioclase 8436T; Sample Weight = 1.0290 gm; J Value = 0.004103 ± 0.000014 (1σ).								
598	-0.00389	0.07278	42.9	0.047	18.3	5.972	0.028	29.9 ± 1.2
701	-0.00922	0.38608	75.4	0.042	13.3	14.161	0.022	37.2 ± 0.5
800	-0.00881	0.52186	84.6	0.025	7.9	13.539	0.039	36.8 ± 2.8
902	-0.00702	0.35402	78.1	0.043	13.0	10.788	0.025	37.8 ± 1.8
1000	-0.00316	0.15549	70.5	0.035	11.8	4.854	0.024	34.0 ± 1.0
1046	-0.00408	0.12308	56.7	0.019	7.0	6.261	0.030	31.2 ± 2.3
1100	-0.00498	0.18570	70.3	0.036	9.9	7.643	0.030	42.5 ± 0.8
1149	-0.00339	0.04634	38.5	0.041	13.7	5.204	0.033	34.3 ± 0.9
1200	-0.00773	0.00902	10.0	0.019	5.1	11.872	0.096	42.7 ± 1.0
							Integrated date	35.5 ± 0.5
Mariscal Mountain Plagioclase 8436TT; Sample Weight = 0.9863 gm; J Value = 0.004103 ± 0.000014 (1σ)								
547	-0.00329	0.04466	40.1	0.051	4.0	5.061	0.025	37.1 ± 0.9
597	-0.00793	0.29602	70.9	0.039	3.0	12.181	0.017	37.8 ± 2.1
650	-0.00920	0.75620	95.0	0.027	1.7	14.129	0.017	47.0 ± 4.3
700	-0.00775	0.46609	83.8	0.019	1.4	11.909	0.026	38.1 ± 2.0
750	-0.00033	0.75095	99.8	0.251	19.3	0.511	0.001	37.5 ± 0.5
800	-0.00044	0.55023	99.3	0.251	18.2	0.669	0.002	40.1 ± 0.4
850	-0.00048	0.46019	98.9	0.253	17.4	0.784	0.002	42.3 ± 0.7
900	-0.00024	0.15443	97.9	0.287	13.1	0.875	0.001	52.0 ± 1.7
950	-0.00018	0.50021	99.7	0.236	11.7	0.269	0.001	58.3 ± 2.1
1000	-0.00232	0.05341	47.9	0.013	1.2	3.570	0.017	30.7 ± 4.2
1050	-0.00255	0.12049	72.5	0.018	1.3	3.920	0.025	40.5 ± 2.5
1100	-0.00421	0.07354	45.0	0.051	4.3	6.467	0.030	34.7 ± 0.4
1150	-0.00523	0.04355	26.3	0.023	2.1	8.027	0.042	30.9 ± 1.9
1175	-0.00782	0.02256	19.6	0.014	0.8	12.014	0.064	51.5 ± 4.0
1200	-0.01024	0.02121	14.9	0.009	0.5	15.733	0.083	47.0 ± 8.3
							Integrated date	43.2 ± 0.4

Paramater $f1 = 1/[1-(37/39)Ca/(37/39)M]$ and $f2 = [1-(36/39)Ca/(36/39)M]$ where ( )Ca is the isotope ratio of argon extracted from irradiated calcium salts and ( )M is the isotope ratio of argon extracted from irradiated unknown; $\%^{40}Ar_R$ is the percent radiogenic ^{40}Ar released at each step; $\%^{39}Ar$ is the percent of the total ^{39}Ar released at each temperature step; apparent age is the date calculated at each temperature step and its error at ± 1σ (this error does not include the error in the J value or the standard error).

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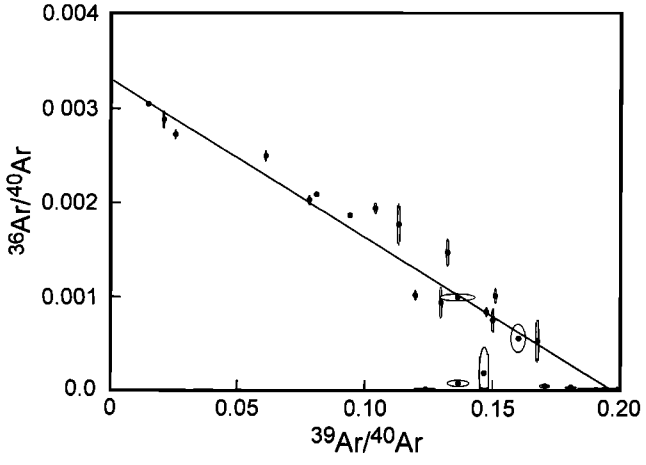


Figure 8. $^{36}Ar/^{40}Ar$ versus $^{39}Ar/^{40}Ar$ inverse correlation diagram for plagioclase samples from the Mariscal Mountain gabbro.

Bend National Park [*Maxwell et al.*, 1967] and in middle Eocene to early Oligocene volcanic rocks elsewhere in the Trans-Pecos region [*Runkel*, 1989]. Although shortening had clearly ceased by this time, continued compressional stresses may have controlled the orientation of dikes emplaced during the Oligocene prior to the onset of Basin and Range extension [*Henry and Price*, 1986].

5.2. Implications for the Magnitude of Vertical Axis Rotations in West Texas

Trans-Pecos Texas, has been variably extended during Basin-Range extension following the early to middle Eocene termination of Laramide contraction. In west Texas, strike-slip displacement occurred along generally west trending faults within and south of the northwest trending Texas Lineament, a poorly defined and controversial structural zone postulated to pass through the Big Bend region [*Stevens and Stevens,* 1989; *Muehlberger*, 1980; *Muehlberger and Dickerson*, 1989] (Figure 1). Paleomagnetic data from several localities within the Big Bend region, including the gabbro sills at Mariscal Mountain anticline, and areas to the south have been used to argue for and against local vertical axis rotations [*Urrutia-Fucugauchi*, 1981; *Sager et al.*, 1992; *Harlan et al.*, 1987]. In the Big Bend Region, *Sager et al.* [1992] argued that significant clockwise vertical axis rotations accompanied right lateral shear along the Texas Lineament (Figure 1), concurrent with Basin and Range extension, although structural studies indicate that such rotations are minor [*Henry*, 1994; also unpublished data, 1995]. We suggest that the paleomagnetic data regarding the presence or absence of vertical axis rotations in the Big Bend region are equivocal, and we briefly review the results of individual studies below. In this analysis we have used the 20-40 Ma reference direction from *Diehl et al.* [1988] and the 30 and 40 Ma reference directions from *Harrison and Lindh* [1982] for middle Tertiary rocks and the Paleocene pole of *Diehl et al.* [1983] for early Tertiary rocks. Discordance parameters for both direction and pole space have been computed using the methods of *Beck* [1980], *Demarest* [1983], and *Beck et al.* [1986] (Table 4). For simplicity, we limit our discussion to results obtained in direction space, although results from direction and pole space are essentially identical. To maintain consistency with the study of *Sager et al.* [1992], only rotation statistics calculated using the expected directions calculated from the *Harrison and Lindh* [1982] 30 and 40 Ma poles for middle Tertiary rocks are discussed below.

As noted previously, paleomagnetic data from the gabbro sills at Mariscal Mountain anticline are consistent with both the expected 20-40 Ma reference direction of *Diehl et al.* [1988] (i.e., $D = 351.1° \pm 2.7°$, $I = 44.2° \pm 3.0°$) and the 40 Ma direction of *Harrison and Lindh* [1982] (i.e., $D = 352.5° \pm 4.0°$, $I = 47.6° \pm 4.1°$) for this area of Texas (Figure 9a). Unfortunately, our data may represent only two or possibly three distinct recordings of the geomagnetic field and thus are of limited value in determining the presence or absence of vertical axis rotations. Regardless, *Sager et al.* [1992] calculated a statistically insignificant rotation of $-4.1° \pm 6.8°$ based on data of *Harlan et al.* [1987]. Calculating discordance values from the group mean direction from this study, we obtain a statistically negligible counterclockwise rotation ($R = -3.9° \pm 6.3°$) and insignificant flattening ($F = -3.3° \pm 4.7°$) (Table 4). As noted by *Sager et al.* [1992], this result is not compelling owing to the small number of independent observations of the geomagnetic field recorded by the gabbro sills at Mariscal Mountain anticline.

Sager et al. [1992] reported paleomagnetic results from a study of 25 sites in lavas of the 27 Ma Rawls Formation of the Bofecillos stratovolcano (Figure 1). Of sites that yielded acceptable results, 19 were of reversed polarity, and one was normal. An in situ mean direction (after inversion of reverse polarity sites to normal polarity) of $D = 15.3°$, $I = 46.9°$ ($k = 13.2$, $\alpha_{95} = 9.4°$) (Table 4; Figure 9b) was reported. A statistically significant clockwise rotation of $21.3° \pm 8.8°$ was obtained, although we calculate $R = 21.3° \pm 11.4°$ and $F = 0.6° \pm 8.2°$ for these same data using the 30 Ma reference direction (Table 4). *Sager et al.'s* [1992] acceptance criteria was quite liberal in that sites with α_{95} values up to 26° were included in their mean direction, and at least two sites used may represent transitional field directions. Recalculating their direction using more stringent criteria (i.e., excluding sites with $\alpha_{95} > 15°$, inclinations $< 20°$, or with virtual geomagnetic poles with latitudes $< 45°$), we obtain a mean of $D = 7.9°$, $I = 46.7°$ ($k = 22.2$, $\alpha_{95} = 9.0°$, $n = 13$ sites) (Figure 9b, Table 4). Comparing this to the 30 Ma reference direction, we also obtain a statistically significant apparent clockwise rotation (Figure 9b) but of diminished magnitude (i.e., $R = 13.9° \pm 10.9°$) (Table 4). We note that only one normal polarity site was reported from the Bofecillos volcano paleomagnetic collection. Because the geomagnetic field from 26 to 30 Ma was of normal polarity approximately 60% of the time, and polarity chrons were of the order of 10 to 400 k.y. (using the Cenozoic timescale of *Berggren et al.* [1985]), we would expect to find a greater number of normal polarity sites if secular variation had been fully averaged. Furthermore, high-precision isotopic dates indicate that most of the Rawls Formation lavas sampled by *Sager et al.* [1992] were erupted during a single pulse at 27.1 Ma [*Henry et al.*, 1994]. Therefore it is questionable whether sufficient secular variation may have been averaged by their data set. Interestingly, secular variation tests performed on the original and recalculated poles indicate that there is significantly more scatter in both data sets than would be expected from normal secular variation based on studies from lavas for the past 0-5 m.y. [*Merrill et al.*, 1991]. The apparent clockwise rotation observed from these rocks may simply result from inadequate sampling of the geomagnetic field or

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Table 4. Summary of Paleomagnetic Data, Poles, and Rotation Statistics for Rocks in the Big Bend Area Used to Argue for the Presence or Absence of Vertical Axis Rotations in Southwest Texas

Study	Rock Unit	D	I	k	α_{95}	n	Plat	Plong	K	A_{95}	R dxn	F	R pole	P
This study	Mariscal Mountain gabbro	348.6°	50.9°	65.1	4.3°	18	80.0°N	185.3°E	42.5	5.4°	-3.9°±6.3° (-2.5°±5.8°)	-3.3°±4.7° (-6.7°±4.2°)	-3.6°±6.0° (-2.3°±5.5°)	-3.1°±5.2° (-5.8°±4.7°)
Sager et al. [1992]	Rawls Formation	15.3°	46.9°	13.2	9.4°	20	76.5°N	348.0°E	13.2	9.4°	21.3°±11.4° (24.3°±11.1°)	0.6°±8.2° (-2.0°±7.8°)	21.3°±9.2° (24.3°±8.8°)	0.5°±8.1° (-1.7°±7.8°)
Sager et al. [1992]*	Rawls Formation	7.9°	46.7°	22.2	9.0°	13	82.3°N	348.2°E	16.8	10.4°	13.9°±10.9° (16.9°±10.6°)	0.8°±7.9° (-1.8°±7.5°)	14.8°±10.1° (17.8°±9.8°)	-0.4°±8.8° (-2.4°±8.5°)
Winters [1967]†‡	Big Bend volcanics	5.8°	36.5°	9.5	13.1°	15	80.2°N	43.5°E	36.5	13.1°	11.8°±13.3° (14.8°±13.1°)	10.9°±10.9° (8.3°±10.6°)	11.7°±11.7° (14.7°±11.5°)	7.4°±10.9° (5.3°±10.7°)
Winters [1967]	Big Bend volcanics	358.4°	39.5°	32.8	8.6°	10	83.6°N	85.3°E	37.6	8.0°	4.4°±9.4° (7.4°±9.1°)	7.9°±7.6° (5.3°±7.2°)	4.9°±7.7° (7.9°±7.3°)	5.4°±7.0° (3.3°±6.7°)
Ryan [1988]	Paisano volcano (Decie Formation)	5.1°	40.1°	10.3	13.0°	14	82.5°N	39.6°E	10.0	13.2°	12.7°±13.9° (14.1°±13.7°)	9.0°±10.8° (5.7°±10.5°)	12.5°±12.0° (14.0°±11.8°)	5.8°±10.9° (3.0°±10.7°)
Ryan [1988]*	Paisano volcano (Decie Formation)	6.4°	42.6°	21.3	17.0°	5	83.1°N	22.5°E	21.0	17.1°	14.0°±18.3° (15.4°±18.1°)	6.5°±13.6° (3.2°±13.3°)	13.8°±15.6° (15.2°±15.4°)	3.9°±14.0° (1.2°±13.8°)
Rapp et al. [1983]	Black Peaks Formation	357.7°	28.1°	25.1	5.9°	25	76.0°N	86.0°E	32.0	5.2°	6.8°±6.2°	24.1°±5.4°	6.8°± 5.3°	17.2°±4.9°
Rapp et al. [1983]*	Black Peaks Formation	355.9°	33.4°	32.9	9.1°	9	78.6°N	97.2°E	31.5	9.3°	5.0°±9.1°	18.8°±7.7°	5.0°±8.4°	14.1°±7.9°

D and I are the declination and inclination of the group mean direction; k is precision parameter [*Fisher*, 1953]; α_{95} is radius of the semi-angle of 95% confidence about the group mean direction; n is number of sites used in the calculation of the mean directions; Plat and Plong are latitude and longitude of the virtual geomagnetic pole (VGP); K is precision parameter calculated for the mean in pole space; A_{95} is radius of the semi-angle of 95% confidence for the VGP; R dxn is rotation estimate calculated in direction space; F is estimate of apparent flattening calculated in direction space; R pole is estimate of apparent rotation calculated in pole space; P is apparent poleward transport. Rotation estimates based on the 20-40 Ma pole of *Diehl et al.* [1988] are shown in parentheses; estimates without parentheses are based on either the 30 Ma or 40 Ma pole of *Harrison and Lindh* [1982] (see text). Rotation estimates for the Black Peaks Formation are based on the Paleocene reference pole of *Diehl et al.* [1983]. The reference poles used are: 30 Ma, 84.7°N, 157.7°E, A_{95} = 3.6°, [*Harrison and Lindh*, 1982]; 40 Ma, 83.4°N, 165.4°E, A_{95} = 3.5° [*Harrison and Lindh*, 1982]; 20-40 Ma, 81.5° N, 147.3°E, A_{95} = 2.4° [*Diehl et al.*, 1988]; and Paleocene, 81.5°N, 192.6°E, A_{95} = 3.2° [*Diehl et al.*, 1983]. VGPs were calculated using the following mean latitude and longitudes, Mariscal Mountain gabbro, 29.1°N, 103.2°W; Rawls Formation volcanic rocks, 29.5°N, 104.0°W; Big Bend volcanic series, 29.5°N, 103.6 °W; Paisano volcano, 30.3°N, 103.75°W; and Black Peaks Formation, 29.4°N, 103.15°W. The poles for the Big Bend volcanic series were calculated using VGPs reported by *Winters* [1967] and *Gilliland and Clark* [1979].

*Denotes our recalculation of the mean direction after exclusion of shallow inclination (< 20°), VGPs < 45°, aberrant directions, and sites with α_{95} values greater than 15°. In cases where statistics for a given parameter were not reported by the original authors, such as pole statistics for the Black Peaks Formation, the reported values are based on our calculations.

†This same data set is also reported by *Gilliland and Clark* [1979].

‡The group mean direction for the Big Bend volcanic series was obtained by inverting Winters [1967] site 12 mean of D = 232.5°, I = 7.2° to obtain a direction whose VGP plotted in the northern hemisphere, i.e. D = 52.5°, I = -7.2°.

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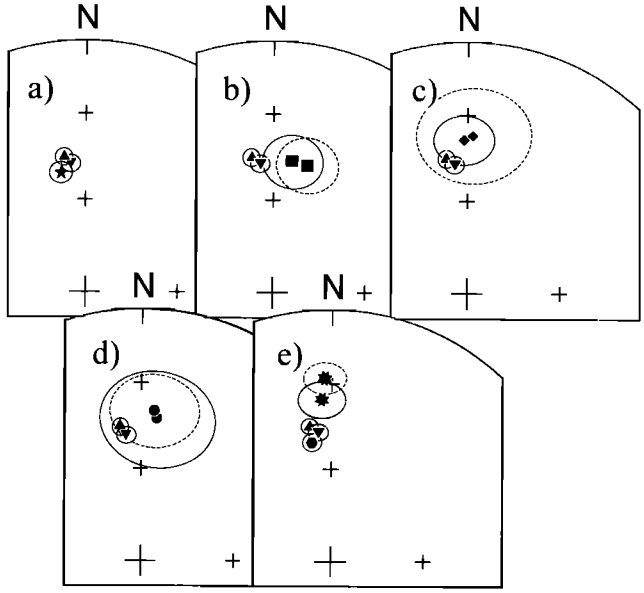


Figure 9. Equal-area projections showing group mean, reference directions, and 95% cones of confidence for various Tertiary rock units in the Big Bend region that have been used to argue for or against local vertical axis rotations: (a) Mariscal Mountain gabbro sills (star) (this study), (b) Rawls Formation (Bofecillos volcano) (square) [*Sager et al.*, 1992], (c) Big Bend volcanics (diamonds) [*Winters*, 1967; *Gilliland and Clark*, 1979], (d) Paisano volcano (circles) [*Ryan*, 1988], and (e) Paleocene/early Eocene Black Peaks Formation (eight-pointed star) [*Rapp et al.*, 1983]. Upward pointing triangle is the expected reference direction for the study area as calculated from the 20-40 Ma pole of *Diehl et al.* [1988]; downward pointing triangle is the expected reference direction calculated from either the 30 Ma or 40 Ma poles of *Harrison and Lindh* [1982] as described in the text; in Figure 9e the octagon is the expected Paleocene direction obtained from the pole of *Diehl et al.* [1983]; Dashed lines for the α_{95} cones for the group mean directions are those calculated from data used in the original studies; solid lines indicate our recalculated or preferred values for the site mean directions. Group mean directions and statistics are listed in Table 4.

from other paleomagnetic causes. Finally, we note that structural evidence from the Bofecillos area demonstrates that total extension within this area is less than 5 to 10% [*Henry*, 1994; also unpublished data, 1995]. It is doubtful that such minimal extension could result in rotations of the magnitude suggested by the paleomagnetic data. Thus, it is unlikely that the apparent rotations recorded by the Rawls Formation rocks are real. We suggest that the paleomagnetic data from Bofecillos volcano do not yield a robust data set in support of clockwise rotation for the Big Bend region.

Winters [1967] and *Gilliland and Clark* [1979] reported paleomagnetic data from Tertiary volcanic rocks of the Big Bend region (Figure 1), including lavas of the Rawls Formation as well as other Eocene and Oligocene volcanic rocks, which we informally call the Big Bend volcanics. They sampled 31 flows, of which 15 gave usable results after AF demagnetization. Not calculated by the original authors, these give a mean direction of D = 5.8°, I = 36.5° (k = 9.5, α_{95} = 13.1°) (Figure 9c; Table 4). *Sager et al.* [1992] reported a statistically significant clockwise rotation of 14.5° ± 6.5° from their analysis of these rocks. In contrast, we obtain insignificant clockwise rotation (R = 11.8° ± 13.3°) and flattening (F = 10.9° ± 10.9°) (Table 4) using the 30 Ma expected direction of *Harrison and Lindh* [1982]. These statistics are misleading, as the mean direction cited here, and as apparently used by *Sager et al.* [1992], includes two sites identified by the original authors as representing transitional field recordings (for example, site 12 with a mean direction of D = 232.5°, I = 7.2° [*Winters*, 1967, Table 5]). Excluding sites with α_{95} > 15° and sites which clearly reflect anomalous directions, and averaging two sites which probably record the same direction [*Winters*, 1967; *Gilliland and Clark*, 1979], we obtain a more appropriate mean of D = 358.4°, I = 39.5° (k = 32.8, α_{95} = 8.6°, n = 10 sites) (Figure 9c; Table 4). In this revised data set, equal numbers of normal and reverse polarity sites are present, the site mean directions pass a class C reversal test [*McFadden and McElhinny*, 1990] at 95% confidence, and the data clearly indicate that some secular variation has been averaged. Calculation of discordance statistics indicates no significant clockwise rotation (R = 4.4° ± 9.4°), although some flattening is suggested by F = 7.9° ± 7.6° (Table 4). This apparent flattening probably results from failure to fully isolate the characteristic magnetization of these rocks or by failure to completely average secular variation. These results are interesting in that several of the sites used in this calculation come from the same area as Rawls Formation data reported by *Sager et al.* [1992] (Figure 1) yet are not consistent with those results. Finally, we note that use of these data in the manner described here, and by *Sager et al.* [1992], is inappropriate because individual sites are distributed over an area of approximately 8500 km^2 (Figure 1) and may span several discrete structural blocks [*Stevens and Stevens*, 1989]. Hence the rotation statistics reported here, and by *Sager et al.* [1992], are of dubious value in determining the sense and magnitude of rotations for this region.

Paleomagnetic results from the 36.3 Ma Paisano volcano (Figure 1) [*Ryan*, 1988; Henry et al., 1994] were reported to yield a clockwise rotation of 12.7° ± 11.5° [*Sager et al.*, 1992]. The mean for 14 independent "cooling units" identified by *Ryan* [1988] is D = 5.1° and I = 40.1° (k = 10.3, α_{95} = 13.0°) (Figure 9d, Table 4). Using the 40 Ma expected direction, we obtain R = 12.7° ± 13.9° and F = 9.0° ± 10.8° (Table 4), neither of which is statistically significant. Examination of the data used by *Ryan* [1988] reveals that most site mean directions were poorly defined (i.e., 55% of the site means had α_{95} > 15°) and that sites with α_{95} values of up to 55° were used to calculate the grand mean direction. Using the same acceptance criteria described above we find only five cooling unit directions which meet minimal acceptance criteria. These give D = 6.4°, I = 42.6° (k = 21.3, α_{95} = 17.0°) (Figure 9d) from which we obtain R = 14.0° ± 18.3° and F = 6.5° ± 13.6° (Table 4). Given the very large α_{95} of this mean, and the small number of independent directions which meet minimal acceptance criteria, the results of this study cannot be given much weight in studies of vertical axis rotations for this area of west Texas.

Finally, paleomagnetic data reported from a magnetostratigraphic study [*Rapp et al.*, 1983] of sedimentary rocks of the Paleocene (Torrejonian)/early Eocene (Clarkforkian) Black Peaks Formation in the northern part of Big Bend National Park (Figure 1) were cited by *Sager et al.* [1992] as supporting clockwise rotation (i.e., R = 9.2° ± 6.1°) for the Big Bend region. Samples for

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this study consisted of mudstones and siltstones collected from 53 (generally 3-4 samples/site) stratigraphic horizons, from which 25 gave usable results. Although some samples were subjected to progressive AF or thermal demagnetization, most samples were demagnetized in only a few steps. Although the results are sufficient for magnetostratigraphy, vector diagrams shown by *Rapp et al.* [1983] do not convincingly demonstrate that characteristic magnetizations were fully isolated. Indeed, most reverse polarity sites are significantly shallower in inclination than normal polarity sites, although these data pass a class C reversal test [*McFadden and McElhinny*, 1990] at 95% confidence. The tendency toward shallow inclinations in reverse polarity sites (typically -8° to -25°) may be caused by failure to adequately remove present-day viscous remanent magnetizations. Overall, the shallow inclinations typical of both normal and reverse polarity sites have probably resulted from compaction shallowing of the remanence vector. In addition to these problems, site mean directions from the Black Peaks Formation rocks are typically poorly determined as 72% of the sites had α_{95} values > 15°. Unfiltered, the paleomagnetic results from the Black Peaks Formation yield a mean direction of $D = 357.7°$, $I = 28.1°$ ($k = 25.1$, $\alpha_{95} = 5.9°$, $n = 3$ normal polarity sites, 22 reverse polarity sites) (Figure 9e, Table 4). As with the previously described studies we excluded sites with inclinations < 20° or VGPs < 45° as they probably represent a failure to isolate the characteristic magnetization or represent transitional field directions. In recognition of the small number of samples collected from most sites we relaxed our acceptance criteria to include sites with $\alpha_{95} \leq 20°$. The resultant mean is $D = 355.9°$, $I = 33.4°$ ($k = 32.9$, $\alpha_{95} = 9.1°$, $n = 9$ sites) (Figure 9e, Table 4). Comparison to the expected Paleocene reference direction of *Diehl et al.* [1988] (i.e., $D = 350.9° \pm 3.8°$, $I = 52.2° \pm 3.4°$) gives $R = 6.8° \pm 6.2°$ and $F = 24.1° \pm 5.4°$ for the original Black Peaks data, or $R = 5.0° \pm 9.1°$ and $F = 18.8° \pm 7.7°$ for the recalculated set (Table 4). In either case the observed clockwise rotations are barely significant or insignificant, although the observed flattening is extreme. Because the rocks have not been tilted significantly or translated northward, it is obvious that this data set, regardless of the selection criteria used to calculate a mean direction, does not provide a faithful recording of the geomagnetic field. Using these data to determine rotation estimates for this region is clearly inappropriate.

Significant problems exist with all the paleomagnetic studies used to argue for or against the existence of vertical axis rotations associated with the Texas Lineament in the Big Bend region. In contrast to the findings of *Sager et al.* [1992], we do not find paleomagnetic evidence for significant clockwise rotations compelling, as nearly all of the estimates can easily be explained by failure to average secular variation or by other paleomagnetic causes. Small (≤ 5° to 10°) clockwise or counterclockwise vertical axis rotations in this region may have occurred during extension or transtension [*Stevens and Stevens*, 1989; *Henry*, 1994] and might be detectable by paleomagnetic means, but paleomagnetic data of higher quality than most of those described here, and which demonstrably average secular variation, are clearly needed in order to resolve this issue.

6. Conclusions

Paleomagnetic data from gabbro sills exposed at Mariscal Mountain anticline demonstrate that sill emplacement postdates Laramide folding. Isotopic dating by the $^{40}Ar/^{39}Ar$ method indicates that sill intrusion occurred at approximately 37 Ma and is thus late Eocene/early Oligocene in age, and not Late Cretaceous or early Eocene as previously argued [*Maxwell et al.*, 1967; *Maxwell and Dietrich*, 1965; *Bumgardner*, 1976]. A posttectonic origin for these intrusions is consistent with an overall cessation of Laramide tectonism prior to late Eocene time, as inferred from geologic relations elsewhere in Trans-Pecos Texas. Our findings indicate that interpretations regarding the relationship between concordant igneous intrusions and deformed strata in other orogenic belts, based solely on field observations and geometry, must be viewed with caution. Finally, we note that paleomagnetic evidence regarding the possibility of local vertical axis rotations from this and other paleomagnetic studies from the Big Bend region is equivocal and does not demonstrate that this region has undergone right-lateral shear in response to Basin and Range extension as postulated by some workers.

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